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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02

Question #: 126

Topic #: 1

[All AWS Certified Developer - Associate DVA-C02 Questions]

A developer is integrating Amazon ElastiCache in an application. The cache will store data from a database. The cached data must populate real-time dashboards.

Which caching strategy will meet these requirements?

A. A read-through cache

B. A write-behind cache

C. A lazy-loading cache

D. A write-through cache

Show Suggested Answer

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by MrTee at *April 23, 2023, 12:25 a.m.*

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Comments

MrTee Highly Voted 2 years, 11 months ago

Selected Answer: D

The best caching strategy for populating real-time dashboards using Amazon ElastiCache would be a write-through caching strategy. In this strategy, when new data is written to the database, it is also written to the cache. This ensures that the most current data is always available in the cache for the real-time dashboards to access, reducing the latency of the data retrieval. Additionally, using a write-through cache ensures that data consistency is maintained between the database and the cache, as any changes to the data are written to both locations simultaneously.

upvoted 17 times

sumanshu Most Recent 1 year, 3 months ago

Selected Answer: D

A) Eliminated

In a read-through cache, when an application tries to retrieve data:
If the data is not available in the cache, the cache retrieves it from the database, stores it in the cache, and returns it to the application.
If the data is available in the cache, it is returned directly.
The cache is updated only when data is requested, which could result in outdated data being served to real-time dashboards if no recent read request has occurred.

upvoted 2 times

sumanshu 1 year, 3 months ago

B) Eliminated

In a write-behind cache, data is written to the cache first, and the database is updated asynchronously (in the background).
This caching strategy prioritizes write performance but risks data loss in the event of a failure because the cache may not yet have updated the database.
For real-time dashboards, there is no guarantee that the cache is always up-to-date because the data is updated asynchronously.

upvoted 1 times

sumanshu 1 year, 3 months ago

C) Eliminated

In a lazy-loading cache, the data is loaded into the cache only when requested.
If the data is not in the cache (a "cache miss"), it is fetched from the database, loaded into the cache, and returned to the application.
While this reduces unnecessary caching, it does not proactively update the cache. Therefore, data in the cache may become stale, which is unsuitable for real-time dashboards.

upvoted 1 times

sumanshu 1 year, 3 months ago

D) Correct
In a write-through cache, every write operation to the database is immediately mirrored in the cache.
This ensures that the cache always contains the most up-to-date data and is ideal for scenarios where real-time consistency is required, such as updating dashboards.

upvoted 1 times

Saudis 1 year, 5 months ago

Selected Answer: D

the keyword => real-time

upvoted 1 times

tirthyakamaldasgupta 1 year, 7 months ago

When using a write-through cache strategy, the cache is updated in real-time alongside the database. This ensures that the cached data remains consistent with the underlying database. According to AWS best practices, this approach pushes data into the cache at the time it is written to the database, reducing the risk of serving stale data.
In contrast, option A (a read-through cache strategy) may result in stale data, particularly if the cache has a time-to-live (TTL) setting that allows data to remain in the cache longer than it remains accurate in the database. This can be problematic for real-time dashboards that require up-to-date information. For these reasons, I opted for option D.

https://aws.amazon.com/caching/best-practices/#:~:text=Write%2Dthrough,also%20pushed%20into%20the%20cache.

Please correct me if my understanding is incorrect, as I am still learning.

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: D

D is the correct answer.

upvoted 1 times

Walker17 2 years, 1 month ago

B. Write Behind Cache.

upvoted 1 times

SerialiDr 2 years, 2 months ago

Selected Answer: D

D. A write-through cache: A write-through caching strategy immediately writes data to both the cache and the database at the same time. This approach ensures that the cache always contains the most recent data, making it highly suitable for applications that require up-to-date information, such as real-time dashboards.

upvoted 2 times

tqiu654 2 years, 4 months ago

Selected Answer: C

ChatGPT:C

upvoted 1 times

Prem28 2 years, 10 months ago

ans- A

Option D, a write-through cache, is incorrect because it would not meet the requirement of populating real-time dashboards. A write-through cache writes data to the cache and the database at the same time. This means that the data in the cache would always be up-to-date, but it

would also mean that the cache would always be lagging behind the database. This would cause a delay in populating real-time dashboards.

upvoted 1 times

loctong 2 years, 10 months ago

Selected Answer: D

A write-through cache strategy involves writing data to both the cache and the underlying database simultaneously. When data is updated or inserted into the database, it is also stored or updated in the cache to ensure that the cache remains up-to-date with the latest data.

upvoted 2 times